Installing Bookstack in Ubuntu 24.04 (WSL)

Explanation of how to install Bookstack in Unbuntu 24.04 WSL

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- Bookstack installation
  - Installing BookStack on Ubuntu/WSL
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- Ways to make available Shelves made in BookStack (installed on WSL) on the web
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objectives

- **Install the WSL ;**
- **Install BookStack;**
- **Set up BookStack.**

WSL

What is WSL

WSL, or **Windows Subsystem for Linux**, is a compatibility layer developed by Microsoft that allows you to run Linux binaries natively on Windows machines. It enables developers to use command-line tools, utilities and Linux applications side by side with Windows applications without the need for a dedicated virtual machine or dual-boot configuration.

WSL has two versions: **WSL 1** and **WSL 2**.

Comparing characteristics

Feature	WSL 1	WSL 2
Integration between Windows and Linux	✅	✅
Fast boot times	✅	✅
Small resource foot print compared to traditional Virtual Machines	✅	✅
Runs with current versions of VMware and VirtualBox	✅	❌
Managed VM	❌	✅
Full Linux Kernel	❌	✅
Full system call compatibility	❌	✅
Performance across OS file systems	✅	❌
systemd support	❌	✅
IPv6 support	✅	✅

WSL can run on Windows 10 version 2004 or later (Build 19041 or later) and on Windows 11.

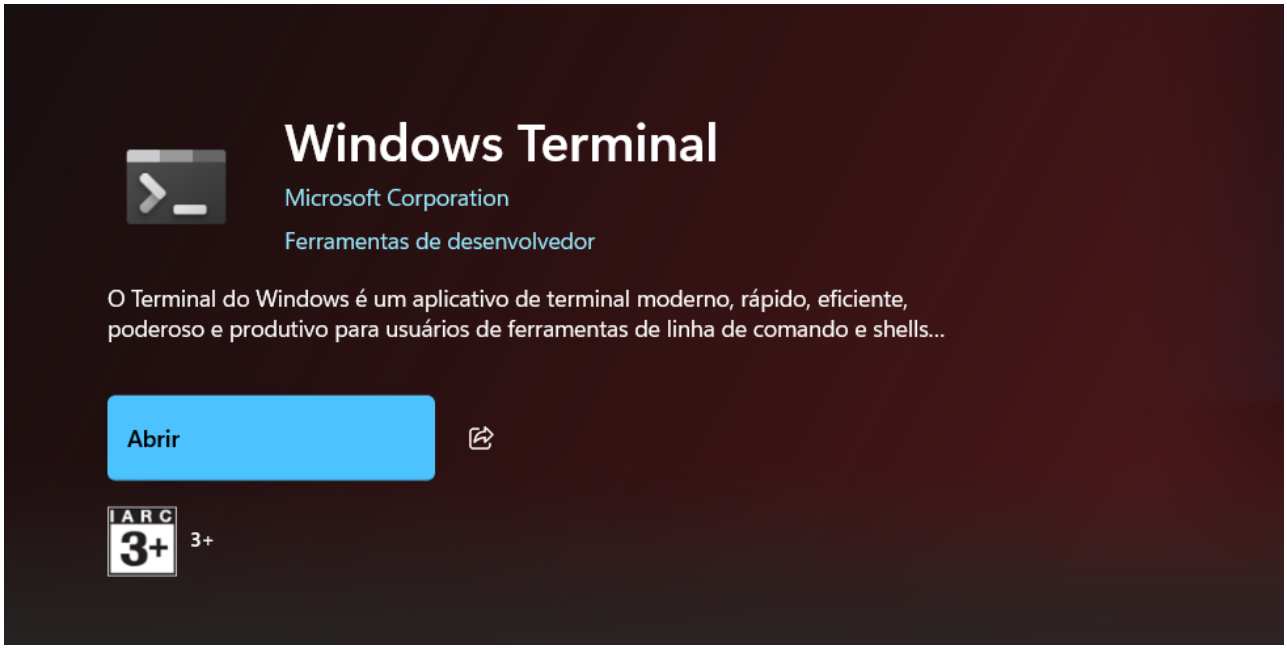
To check the version and build of Windows:

1. Press **Windows + R**
2. type **winver**
3. Check the version displayed in the window.

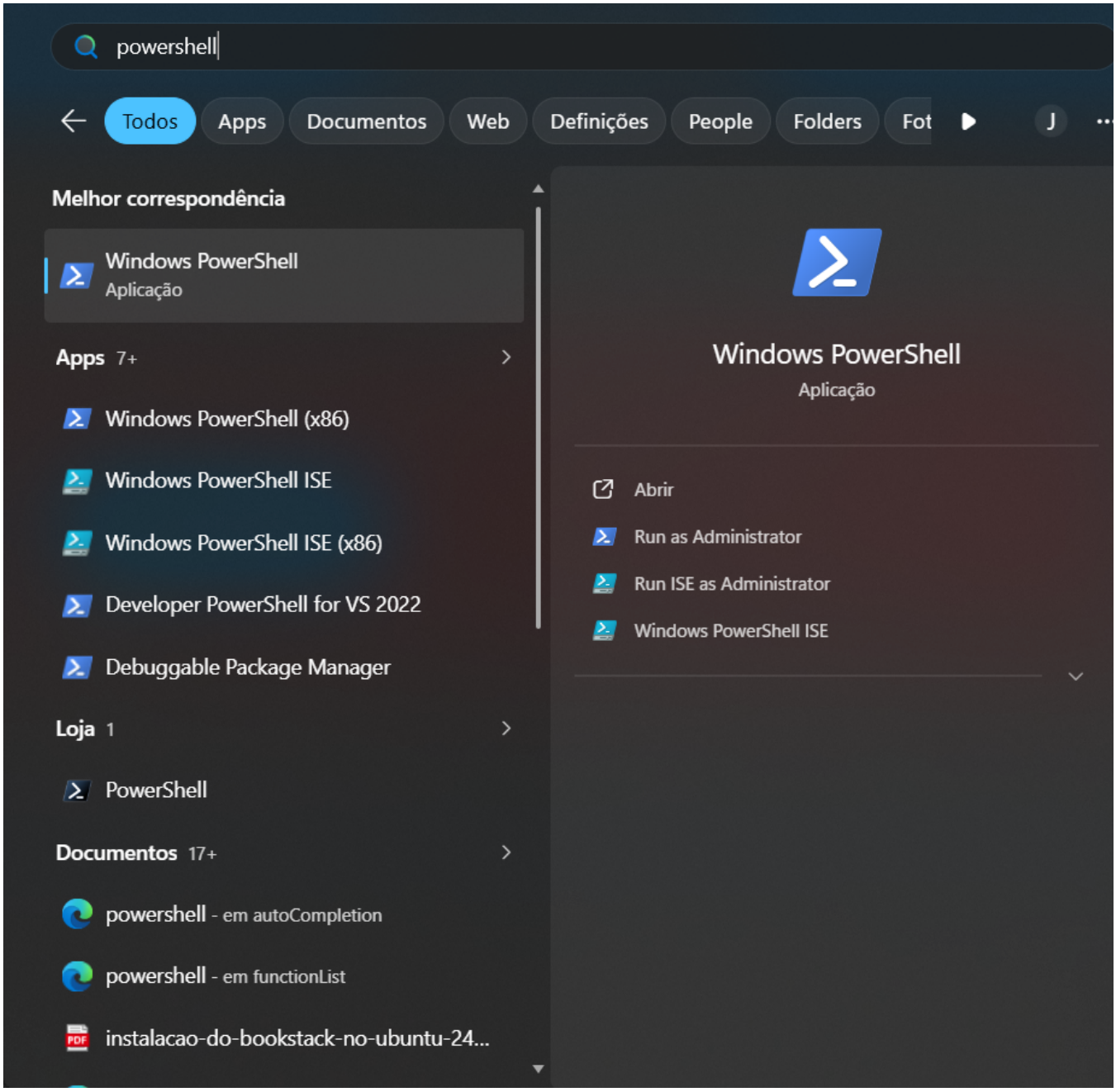

Acerca do Windows
Windows 11
Microsoft Windows
Versão 25H2 (Compilação do SO 26200.8894)
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Installation of WSL

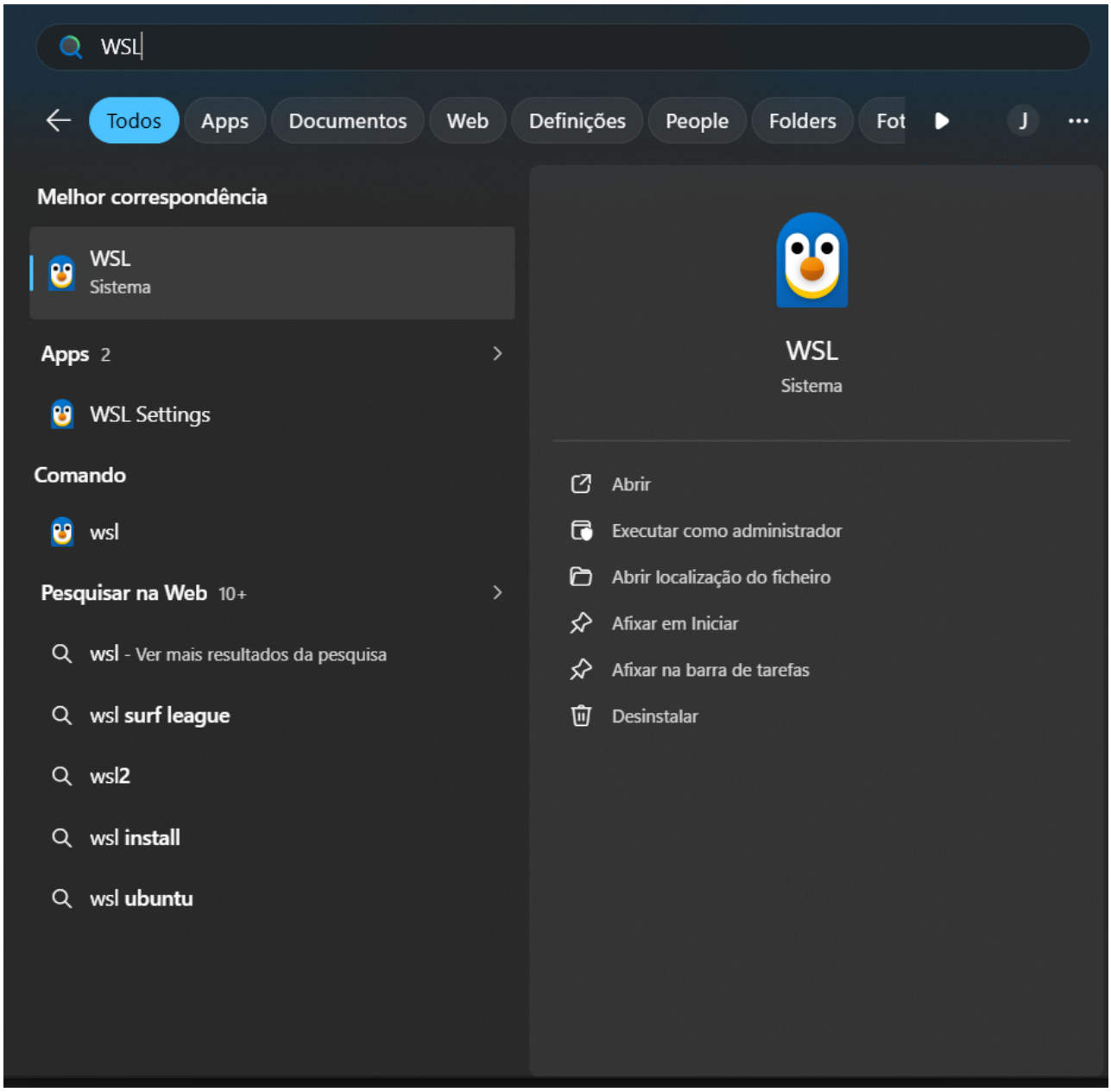
1. From the Microsoft Store, install Windows Terminal.



2. Open PowerShell as an Administrator. Right-click on the icon of PowerShell → Run as administrator.



3. Run the command to install WSL:
4. Restart the computer after installation
5. After restarting, search for "WSL" in the Windows search bar and open the application.



6. The WSL may take a few minutes to start, it is normal for the screen to go unresponsive for some time.
7. When the initial prompt appears, set up the system:

- Choose a username
- Set a password

This same password will be used for administrative tasks within Ubuntu (such as the comando sudo).

Bookstack installation

Installing BookStack on Ubuntu/WSL

Practical rules for choosing ports in WSL/Windows

Any network port must be an integer between 1 and 65535.

Classification of the doors:
Privileged doors: 1-1023
User ports: 1024-49151
Dynamic/ephemeral doors: 49152-65535

For this example, port 8088 will be used.

1. Update of two System Packages

Update the repositories and all packages installed in Ubuntu:

```
sudo apt update && sudo apt upgrade -y
```

2. Installation of System Dependencies (PHP, MariaDB, Nginx and Utilities)

2.1 PHP Installation and Required Extensions

Install PHP-FPM and all required extensions for BookStack:

```
sudo apt install -y \
    php-fpm \
    php-cli \
```

```
    php-mbstring \
    php-xml \
    php-mysql \
    php-common \
    php-curl \
    php-zip \
    php-gd \
    php-bcmath \
    php-tokenizer \
    git \
    unzip \
    curl \
    openssl
```

Check the installed version of PHP and note this information.

```
php -v | head -n 1
```

> Note: If the version has three numbers (e.g. PHP 8.3.5), use only the first two when configuring services and Nginx (e.g. PHP 8.3).

3. Installation of Nginx and MariaDB Server

```
sudo apt install -y nginx mariadb-server
```

4. Service Initialization in WSL

> Note: Replace php8.3 with the version installed on your system. Example: if you are using PHP 8.5, run sudo service php8.5-fpm start.

```
sudo service mariadb start
sudo service php8.3-fpm start
sudo service nginx start
```

5. MariaDB Database Configuration

5.1 Database creation and user

Access the MariaDB console as root:

```
sudo mysql
```

Execute the SQL commands (replace your password with the desired password):

```
CREATE DATABASE bookstack;
CREATE USER 'bookstackuser'@'localhost' IDENTIFIED BY 'sua_senha';
GRANT ALL PRIVILEGES ON bookstack.* TO 'bookstackuser'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

6. Composer installation and BookStack download

6.1 Install Composer

```
curl -sS https://getcomposer.org/installer | php
sudo mv composer.phar /usr/local/bin/composer
```

6.2 Download BookStack via Git

Clone the latest stable version:

```
cd /var/www
sudo git clone \
    https://github.com/BookStackApp/BookStack.git \
    --branch release \
    --single-branch \
    bookstack
cd /var/www/bookstack
```

6.3 Installing PHP Dependencies

```
sudo composer install --no-dev
```



Accept (yes) and click enter.

7. BookStack App configuration

7.1 Environment File Configuration (.env)

Copy the sample file:

```
sudo cp .env.example .env
sudo nano .env
```

Adjust the line:

```
APP_URL=http://localhost:8088
```

Copy and paste after the APP_URL line :

```
SESSION_SECURE_COOKIE=false
SESSION_SAME_SITE=lax
COOKIE_DOMAIN=
```

Adjust the lines :

```

# Database details
DB_DATABASE=bookstack
DB_USER=bookstackuser
DB_PASSWORD=sua_senha (A mesma senha definida na base de dados )
```

At the end of the paste file

```
SESSION_DRIVER=database
SESSION_SECURE_COOKIE=false
SESSION_SAME_SITE=lax
```

Save with **Ctrl+O**, Enter, and exit with **Ctrl+X**.

7.2 Application Key Generation and Table Migration

```
sudo php artisan key:generate --force
sudo php artisan migrate --no-interaction --force
```

8. Nginx Web Server configuration

8.1 Creating the Virtual Host

Open the file:

```
sudo nano /etc/nginx/sites-available/bookstack
```

> Note: Replace php8.3 with the installed version.

Example: `fastcgi_pass unix:/run/php/php8.5-fpm.sock;` if ph8.5 is the installed version.

Paste the content below:

```
server {
    listen 8088;
    server_name localhost;
    root /var/www/bookstack/public;

    index index.php index.html;

    location / {
        try_files $uri $uri/ /index.php?$query_string;
    }

    location ~ \.php$ {
        include fastcgi_params;
        fastcgi_pass unix:/run/php/php8.3-fpm.sock;
        fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
    }
}
```

Save and close (**Ctrl+O**, Enter, **Ctrl+X**).

9. Site Activation and Nginx Reboot

```
sudo ln -s /etc/nginx/sites-available/bookstack /etc/nginx/sites-enabled/
sudo rm -f /etc/nginx/sites-enabled/default
sudo nginx -t
sudo service nginx restart
```

10. Permissions and Final Validation

10.1 Adjust Permissions

```
sudo chown -R www-data:www-data /var/www/bookstack
sudo chmod -R 755 /var/www/bookstack
sudo chmod -R 775 /var/www/bookstack/storage /var/www/bookstack/bootstrap/cache
/var/www/bookstack/public/uploads
```

10.2 Cache Clean and Test

```
cd /var/www/bookstack
sudo php artisan config:clear
sudo php artisan cache:clear
curl -kI http://localhost:8088
```

The expected return is:

- **HTTP/1.1 200 OK** running application **OR**
- **HTTP/1.1 302 Found** active HTTPS redirect

11. Maintenance and Troubleshooting Guide in WSL

11.1 Quick Boot after Restarting Windows

Whenever you turn off or restart your computer, WSL services will be interrupted. To reconnect BookStack, simply open the Ubuntu terminal and run this unique command:

Note: Replace `php8.3` by the installed version.

```
sudo service mariadb start && sudo service php8.3-fpm start && sudo service nginx restart
```

11.2 Useful Diagnostic Commands

If the page shows any unexpected behavior in the future, run the following commands to identify the exact reason:

Check the BookStack error log:

```
tail -n 50 /var/www/bookstack/storage/logs/laravel.log
```

Check the Nginx error log:

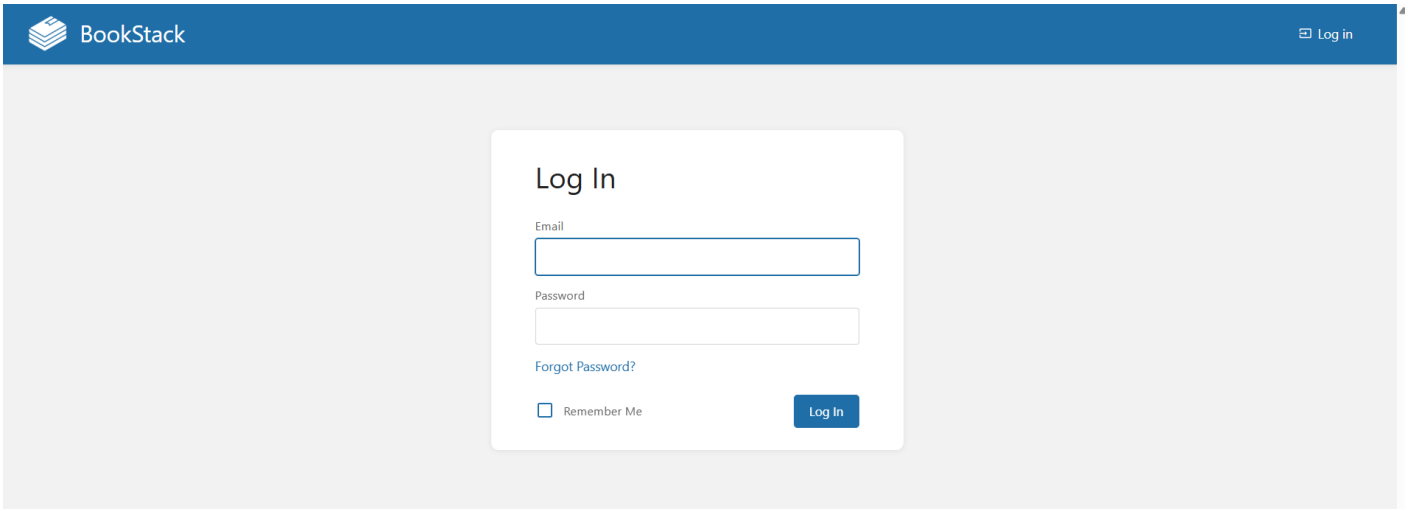
```
sudo tail -n 50 /var/log/nginx/error.log
```

Open BookStack in the browser

After completing all the previous processes (WSL, Ubuntu, PHP, MariaDB, BookStack and Nginx), open a browser and go to:

```
http://localhost:8088
```

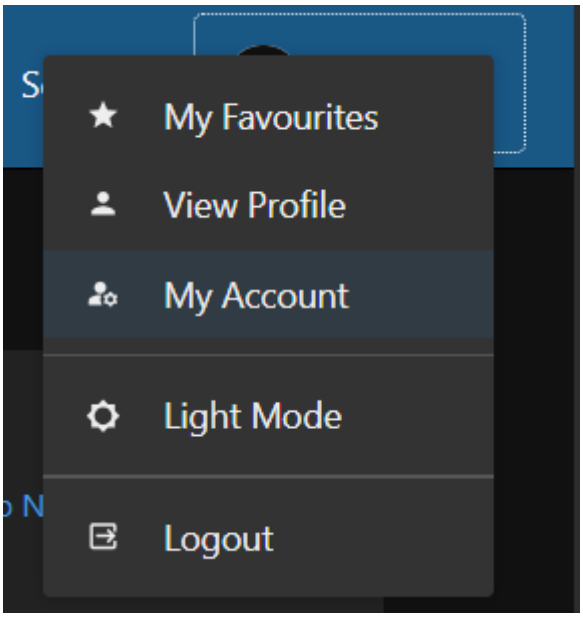
Standard BookStack credentials



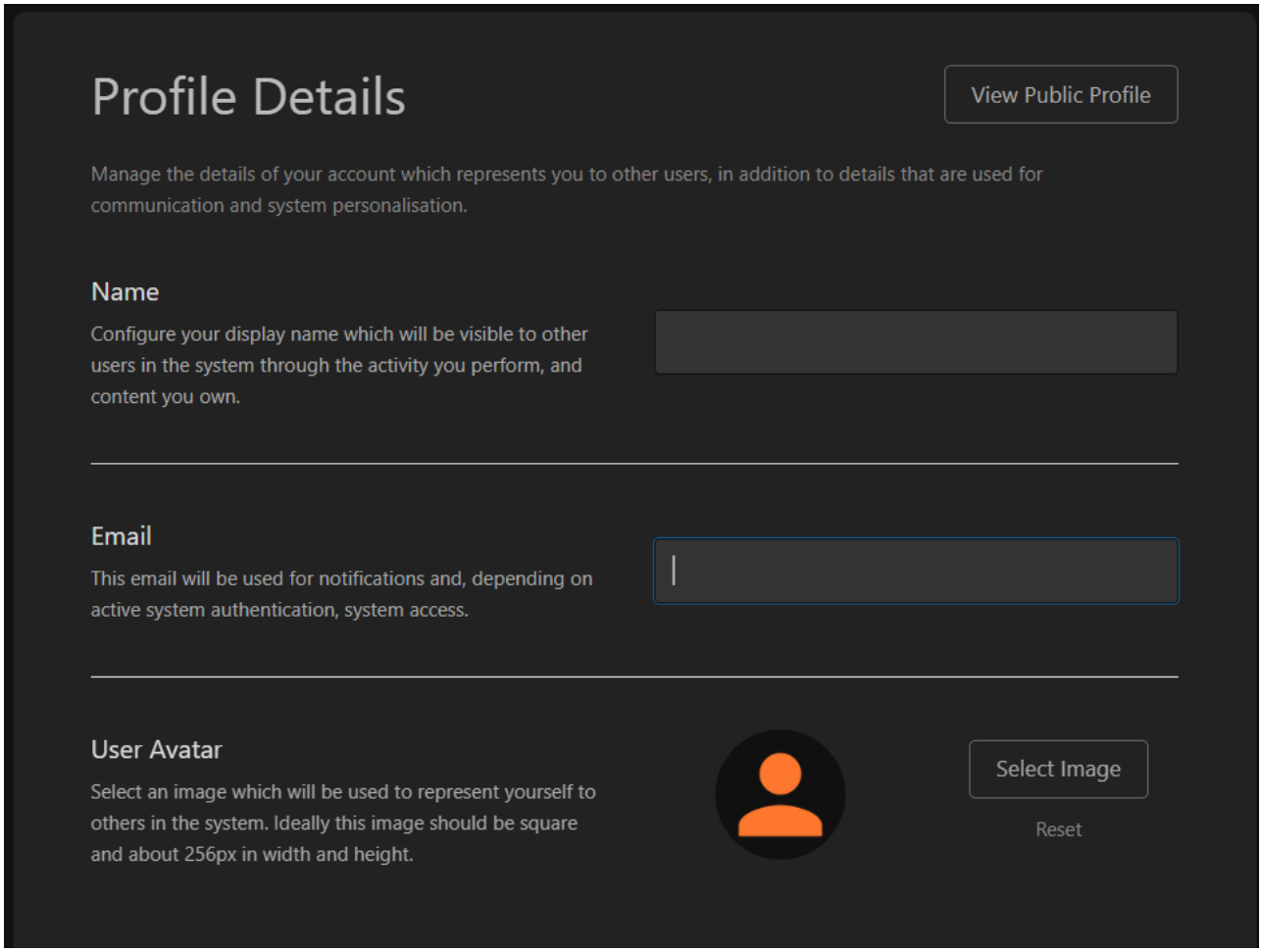
Use the initial credentials:

- **E-mail:** admin@admin.com
- **Password:** password

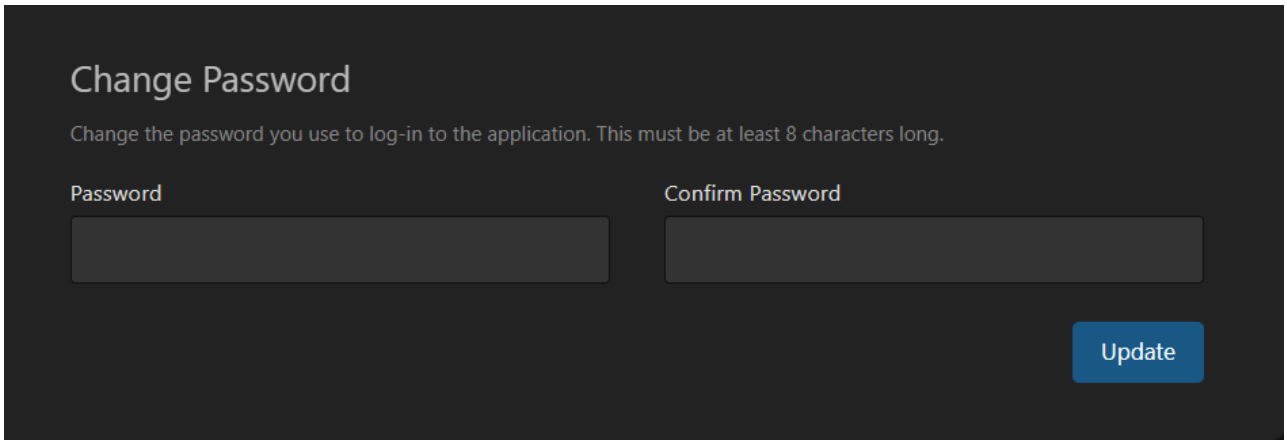
Once you log in, go to the **User** tab and change:



- Username and email



- The password to ensure security.



Important WSL tip

If you close the Ubuntu (**WSL**) terminal window or restart **Windows**, **MariaDB**, **PHP-FPM** and **Nginx** services stop automatically.

Note: Replace `php8.3` by the installed version.

Before trying to access BookStack again, run:

```
sudo service mariadb start && sudo service php8.3-fpm start && sudo service nginx start
```

Ways to make available Shelves made in BookStack (installed on WSL) on the web

3 ways to put your BookStack shelves on the web

Next, I will present three ways to put your BookStack shelves on the web. In this page, I mention only how this can be done without going into too many details of each configuration but providing the basics. I also include the cost of each option, a small comparison and a recommendation if you intend to use them.

1. **VPS in the Cloud (Virtual Private Server)**

This is the most traditional and professional approach. It consists of renting a Linux machine (usually Ubuntu) running 24/7 at a cloud provider such as Hetzner, DigitalOcean, Linode or AWS.

How it is done:

1. Contracting: A Linux instance (VPS) is created at a cloud provider.
2. Domain Pointing: In your domain registrar (e.g. Cloudflare, GoDaddy), an A Record is created by pointing your domain (suadocumentacao.com) to the VPS public IP.
3. Installation: Run the official BookStack script on VPS's Ubuntu.
4. Security (SSL): The certbot tool (Let's Encrypt) is used to generate a free and automatic HTTPS certificate.
5. Migration: Export the WSL database (mysqldump) and the images/uploads folder, importing everything into the VPS.

Advantages:

- High stability and availability (99.9%);
- Total control over the machine;
- Custom domain support;
- Your personal computer does not need to stay on.

Disadvantages:

- Requires server maintenance (Ubuntu updates, firewall rules, security);
- Requires initial data migration configuration.

Cost:

- VPS: $4 to $16 USD/month (inbound machines on Hetzner or DigitalOcean).
- Domain: ~$10 to $15 USD/year (optional but recommended).
- SSL Certificate: Free (via Let’s Encrypt).

2. Cloudflare Tunnel on WSL

An ideal solution for testing or temporary sharing. It creates an "encrypted tunnel" between the application running on your local WSL and Cloudflare’s global servers, exposing your website to the internet without opening ports on your home router.

How it is done:

1. Installation: The Cloudflare client (cloudflared) is installed directly in your WSL terminal.
2. Tunnel Opening: Execute the command cloudflared tunnel --url https://localhost:8888 --no-tls-verify.
3. Access: Cloudflare instantly generates a secure public link (e.g.: https://seu-nome-123.trycloudflare.com).
4. Visibility: In the BookStack dashboard, "Allow Public Access" is enabled so that visitors can read the book without having to log in.

Advantages and Disadvantages

Advantages:

- 100% free;
- Quick setup (less than 5 minutes);
- Hides your home IP;
- Provides factory-safe, automatic HTTPS without needing manual certificates.

Disadvantages:

- The website only goes online if your computer is connected and with WSL/terminal open; if the PC shuts down or hibernates, the documentation goes out of the air; the free link changes if the command is restarted (unless configured with a own domain at Cloudflare).

cost:

- 100% Free.

3. **PaaS/Container providers (Render, Fly.io)**

Platforms as a Service (PaaS) manage the entire infrastructure for you. Instead of managing an entire Linux server, you provide BookStack as a Docker container, and the platform takes care of

hosting, database, and security.

How it is done:

1. Account creation: An account is created on the platform (Render, Fly.io or similar).
2. Deploy the container: Get the official image of BookStack's Docker (solidnerd/bookstack or lscr.io/linuxserver/bookstack).
3. Managed Database: Connect the container to a MySQL/PostgreSQL database provided by the platform itself.
4. Media Configuration: Because containers are "volatile" (they can be restarted), BookStack is configured to save uploaded images in an external cloud storage (e.g. Amazon S3 or Cloudflare R2).

Advantages and Disadvantages

Advantages:

- Zero operating system or Nginx maintenance;
- Website available 24/7 without relying on your computer;
- Automatic SSL and simple integration with domains.

Disadvantages:

- Requires understanding concepts of containers and environment variables;
- Storing uploads/images requires setting up an external file service (S3/R2).

cost:

- Entry / Trial Plan: Free or with initial credits (there may be downtime/hibernation limitations on 100% free plans).
- Continuous Use (Production): From $5 to $7 USD/month to keep the application and database running without interruptions.

comparative table

Criterion	1. Cloud VPS	2. Cloudflare Tunnel (WSL)	3. PaaS providers (Render / Fly.io)
Availability	24/7 (Always online)	Depends on your PC (Falls if you power down)	24/7 (Always online)
Estimated cost	$4 - $16 / monthly	**Free ($0)**	Free (testing) at $5-$7/monthly (production)
Setup difficulty	Medium (requires Linux commands)	Very Low (2 commands)	Medium / High (requires Docker/variables)

Criterion	1. Cloud VPS	2. Cloudflare Tunnel (WSL)	3. PaaS providers (Render / Fly.io)
Server maintenance	Yes (you handle Ubuntu/Nginx)	No	**No** (managed by the platform)
SSL certificate (HTTPS)	Free (via Certbot)	**Automatic**	**Automatic**
Suitable for:	Final projects / Companies	Rapid tests / Demonstrations	Who wants website 24/7 without managing server

Summary of the Recommendation

- If you just want to show your book for a short time without spending anything: Choose Option 2 (Cloudflare Tunnel).
- If you want a professional, definitive and cheap documentation: Choose Option 1 (VPS in the Cloud).
- If you want the website running 24/7 but don’t want to learn how to manage Linux servers: Choose Option 3 (PaaS/Render).

Conclusion

Installing BookStack on WSL demonstrates how you can integrate Linux tools into your Windows environment in an efficient and organized way. Throughout this process, essential components such as Ubuntu, PHP, MariaDB, Composer and Nginx have been configured, ensuring that the application runs correctly. After adjusting permissions, performing migrations and validating server configuration, BookStack becomes fully operational and accessible via browser. With this, the user has a functional platform for documentation management, ready to be customized and used in daily life.